

Homework - 5

1) Given $X = [X_1, X_2, X_3, X_4]^T$ with

covariance matrix $\Sigma = \begin{bmatrix} 0.71 & -0.43 & 0.43 & 0 \\ -0.43 & 0.46 & -0.26 & 0 \\ 0.43 & -0.26 & 0.46 & 0 \\ 0 & 0 & 0 & 0.2 \end{bmatrix}$, $Q = \Sigma^{-1} = \begin{bmatrix} 5 & 3 & -3 \\ 3 & 5 & 0 \\ -3 & 0 & 5 \\ 0 & 0 & 0 \end{bmatrix}$

(a) Are X_3 and X_4 correlated?

Ans NO. They are not correlated, since $\text{Cov}(X_3, X_4) = 0$

(b) Are X_3 and X_4 conditionally correlated given the other variables?

Ans NO. X_3 and X_4 are conditionally independent (uncorrelated) given the others. This is because, conditional correlations are read from the precision matrix Q , if $Q_{ij} = 0$, then X_i and X_j are conditionally independent given all other variables.

(c) Markov blanket of X_i includes all nodes directly connected to it in the precision matrix.

Row 2 of $Q = [3, 5, 0, 0]$.

Nonzero entries correspond to X_1 and X_2 itself.

Ans Markov blanket of X_2 : $\{X_1\}$, because only $Q_{21} = 3$ is non zero (besides diagonal).

(d) Covariance matrix of $Y = [Y_1, Y_2]^T$

Given $Y_1 = X_1 + X_4$, $Y_2 = X_2 - X_4$

We can write $Y = AX$, where

$$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

Then

$$\Sigma_y = A \Sigma A^T$$

$$\text{Var}(Y_1) = \text{Var}(X_1 + X_2)$$

$$= \text{Var}(X_1) + \text{Var}(X_2) + 2\text{Cov}(X_1, X_2)$$

$$= 0.71 + 0.2 + 2(0)$$

$$= 0.91$$

$$\text{Var}(Y_2) = \text{Var}(X_2 - X_4)$$

$$= \text{Var}(X_2) + \text{Var}(X_4) - 2\text{Cov}(X_4, X_2)$$

$$= 0.46 + 0.2 - 0$$

$$= 0.66$$

$$\text{Cov}(Y_1, Y_2) = \text{Cov}(X_1 + X_4, X_2 - X_4)$$

$$= \text{Cov}(X_1, X_2) - \text{Cov}(X_1, X_4) + \text{Cov}(X_4, X_2) - \text{Cov}(X_4, X_4)$$

$$= -0.43 - 0 + 0 - 0.2$$

$$= -0.63$$

∴

$$\text{Covariance matrix } \Sigma_y = \begin{bmatrix} 0.91 & -0.63 \\ -0.63 & 0.66 \end{bmatrix}$$

②

$$E\text{-Step: } r_{ik} = \frac{\frac{1}{2} \phi(x_i | \mu_k, 1)}{\sum_{j=1}^2 \frac{1}{2} \phi(x_i | \mu_j, 1)}$$

$$M\text{-Step } \mu_k^{\text{new}} = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}$$

Dataset $x^{(1)} = -1$ $x^{(2)} = 1$

⑥ One EM update from $\mu_1 = -2, \mu_2 = 2$

For $x = -1$

$$r_{11} = \frac{\phi(-1 | -2, 1)}{\phi(-1 | -2, 1) + \phi(-1 | 2, 1)} = \frac{e^{-\frac{1^2}{2}}}{e^{-\frac{1^2}{2}} + e^{-\frac{3^2}{2}}} = \frac{1}{1 + e^{-4}} \approx 0.98$$

$$\text{so } r_{12} = 0.02$$

For $x = 1$: $r_{22} \approx 0.98, r_{21} \approx 0.02$ as the symmetry

M-Step:

$$\mu_1^{\text{new}} = \frac{0.98(-1) + 0.02(1)}{0.98 + 0.02} \approx -0.964$$

$$\mu_2^{\text{new}} = \frac{0.02(-1) + 0.98(1)}{0.02 + 0.98} \approx 0.964$$

⑦ Will EM Converge to $\mu = -1, \mu_2 = 1$

No. with symmetric data $\{-1, 1\}$, equal weights, and equal variances, the EM updates preserve symmetry: if $\mu_1 = -\mu_2$

then after one E-step / M-step one can show

$$m_{\text{new}} = 2\sigma(2m) - 1 = \tanh(m)$$

so the fixed points solve $m = \tanh(m)$, whose only real solution is $m=0$. Starting $(\mu_1, \mu_2) = (-2, 2)$ then sequence $m_{t+1} = \tanh(m_t)$

$m_{t+1} = \tanh(m_t)$ decreases to 0, hence $(\mu_1, \mu_2) \rightarrow (0, 0)$, not to $(-1, 1)$.

Answer: No; EM Converges to $(0, 0)$

(C) What's the fixed point when initialized $\mu_1 = \mu_2 = 2$

If $\mu_1 = \mu_2$, responsibilities are $r_{ik} = 1/2$ for all i, k . Then M-step sets each mean to overall sample mean, which is 0 for $\{-1, 1\}$. Once both means are 0, they stay 0.

Answer: $(\mu_1, \mu_2) = (0, 0)$ is the fixed point

(d) Fixed point of k-means from $\mu_1 = -2, \mu_2 = 2$

Assignments: $-1 \rightarrow$ cluster at -2 ; $1 \rightarrow$ cluster at 2

$$\mu_1 \leftarrow -1, \mu_2 \leftarrow 1$$

They are now the exact means of their (single point) clusters, so assignment don't change

Answer: k-means converges to $(\mu_1, \mu_2) = (-1, 1)$